Shop Sergeant

System Requirements Review

Jeff Irland · April 30, 2026 · Makersmiths Leesburg (MSL)

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Why We Need This

Makersmiths' current paper-only work task volunteering process has several weaknesses. A new automated process, called **Shop Sergeant**, is proposed to remove much of the friction in the current process, make work task activity more visible, and gather the data for our non-profit status. The process uses Slack as its primary communication/control medium, but can also be trigger via a few clicks on a mobile phone web form.

Current paper-only process has 5 critical gaps:

Gap	Problem	Impact
Incomplete records	Members rarely write their name and date after completing a task	No reliable record of who did what — compliance data simply doesn't exist
No cumulative history	Sign-up sheets are discarded when replaced	No historical data to support Makersmiths' 501©(3) volunteer-labor documentation
No steward monitoring	Stewards have no way to see overdue or ignored tasks	Problems surface only when equipment breaks or a safety issue arises — never proactively
No scheduling	Periodic tasks have no mechanism to track whether they were done on time	Maintenance gets forgotten; equipment degrades; safety risks accumulate
No recognition	Top contributors can't be identified because records don't exist	Highly engaged members receive no recognition; motivation and community goodwill suffer

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Actors — Human

Ellipse shape in all diagrams

Actor	Role Summary
Member	Completes tasks; logs completions via QR or Slack; queries own history
Steward	Manages tasks for their location; generates sign-up sheets; monitors status
Shop Steward	Oversees entire shop; coordinates stewards; escalates issues
Shop Sergeant	Manages the process; monitors KPIs; full system access

Actors — System

Rectangle shape in all diagrams

Actor	Role Summary
Physical Sign-Up Sheet	Shop-floor touchpoint — 8.5×11" landscape, posted per location
QR Code	Encodes mobile web form URL; triggers digital logging
Mobile Phone	Member's own device — no app install required
Mobile Web Form	≤ 3-tap completion UI opened via QR scan
Slack	Primary interface — all notifications and commands
Google Sheets	**Single source of truth** for all data
Claude AI Model	Natural language query + Slack Update task processing
APScheduler	Time-based automation — reminders and scheduled reports
Agentic Workers	Python Slack bot — mediates all 4 pipelines

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Technology Stack

Component	Technology	Role
Bot framework	Python 3.11+ · Slack Bolt	Listens to Slack events; dispatches commands to the appropriate pipeline
AI agent	Claude API (`claude-sonnet-4-6`)	Natural language query handling; Slack Update task completion processing
Data store	Google Sheets API v4	Authoritative storage for all tasks and completion records
Sheet generation	Jinja2 + wkhtmltopdf	Renders HTML sign-up sheet templates to PDF for printing
QR codes	`qrcode[pil]` Python library	Generates QR codes embedded in sign-up sheets
Scheduling	APScheduler	Sends recurring reminders; monitors overdue periodic tasks
Testing	pytest	Unit and integration tests for all pipeline components

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Agentic AI Capabilities

The bot uses `claude-sonnet-4-6` for two primary capabilities:

1. Natural Language Query Processing

- Members & Stewards ask questions in plain English:

> "What tasks are still open in the metalshop?"
> "How many tasks has @alex done this month?"

- Claude parses the intent, constructs the Google Sheets query, and formats the response inline in Slack.

2. Slack Update Task Processing

- Member posts in `#shop-queries` :

> "I just finished cleaning the metalshop floor"

- Claude identifies the most likely task, presents it for member confirmation, then writes to Google Sheets.

What Claude does NOT do:

- Modify data without member confirmation
- Execute admin commands from `#shop-queries`
- Access data outside Google Sheets

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Task Database Pipeline

Maintains the master list of all tasks in Google Sheets; supports add, update, and delete

Who can modify the task catalog?

- **Steward** — own location only
- **Shop Steward** — all locations
- **Shop Sergeant** — all locations + system config

Task ID format: `SHOP-LOCATION-NNN` e.g., `MSL-METAL-003` = Metalshop task #3

Every task requires: unique Task ID · name · frequency · location

Validation rules:

- Reject duplicate Task IDs
- Reject duplicate task names within same location

Key Requirements

#	Requirement
DB-1	Google Sheets = single source of truth
DB-2	YAML bulk-load via CLI script
DB-3	Add / update / delete via Slack in `#shop-admin`
DB-4	Only Steward (own location), Shop Steward, Sergeant
DB-5	Task ID + name + frequency + location required
DB-6	Reject duplicate IDs and names within location
DB-7	Every change notifies `#shop-alerts`

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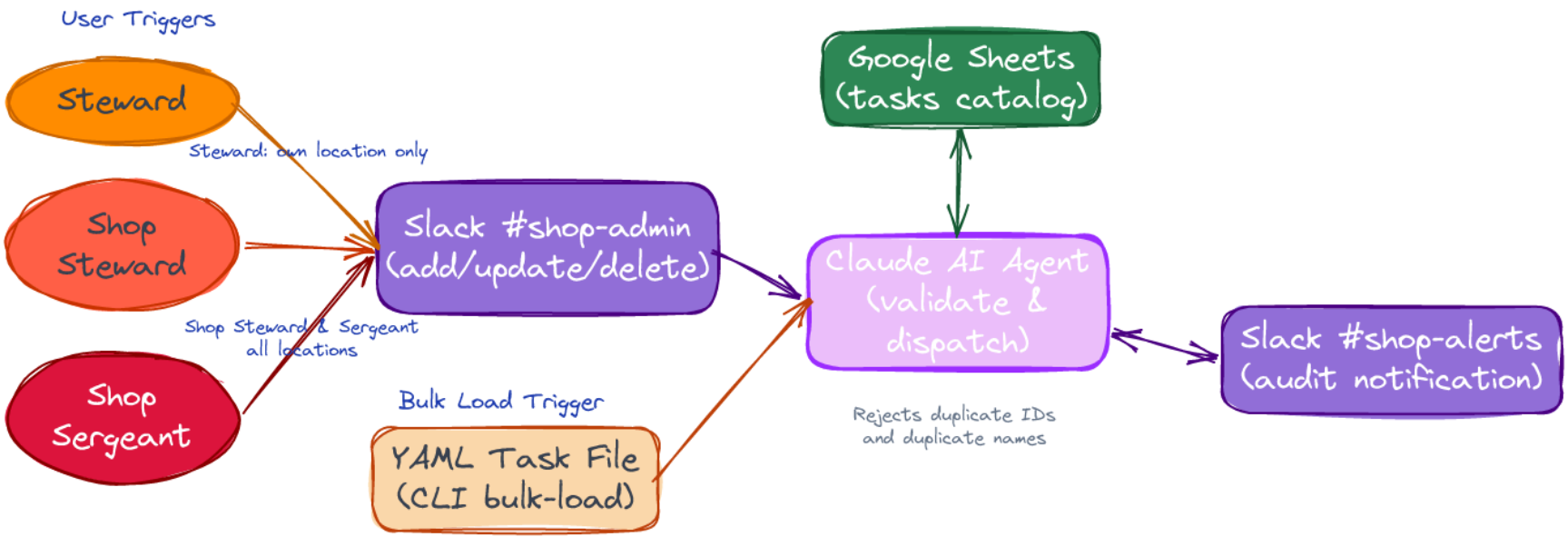
Task Database Pipeline



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Task Sheet Pipeline

Generates printable sign-up sheets from the task database and posts them around the shop

Specification:

- 8.5 × 11 inches, landscape orientation
- One sheet per location · Legible in black and white
- Makersmiths logo in header

Required columns:

Column	Content
Task Name	Human-readable description
Frequency	Weekly, Monthly, Quarterly, etc.
Supervisor Required	Yes / No
Completion Date	Blank — member writes by hand
Member Name	Blank — member writes by hand

QR Code: Single code in page header (one per location)

Task Display Rules

- Tasks grouped by location
- TBD / empty tasks silently omitted
- Ordered by frequency (most frequent first)

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Task Sheet Pipeline — Key Requirements

#	Requirement
TS-1	Generate for single location or all locations
TS-2	Authorized roles only (Steward = own location)
TS-3	Via Slack command or CLI
TS-4	`#shop-bulletin` notified when sheet is ready
TS-5	`#shop-alerts` notified each generation
TS-6	Bot reminds stewards when task changes make a refresh advisable

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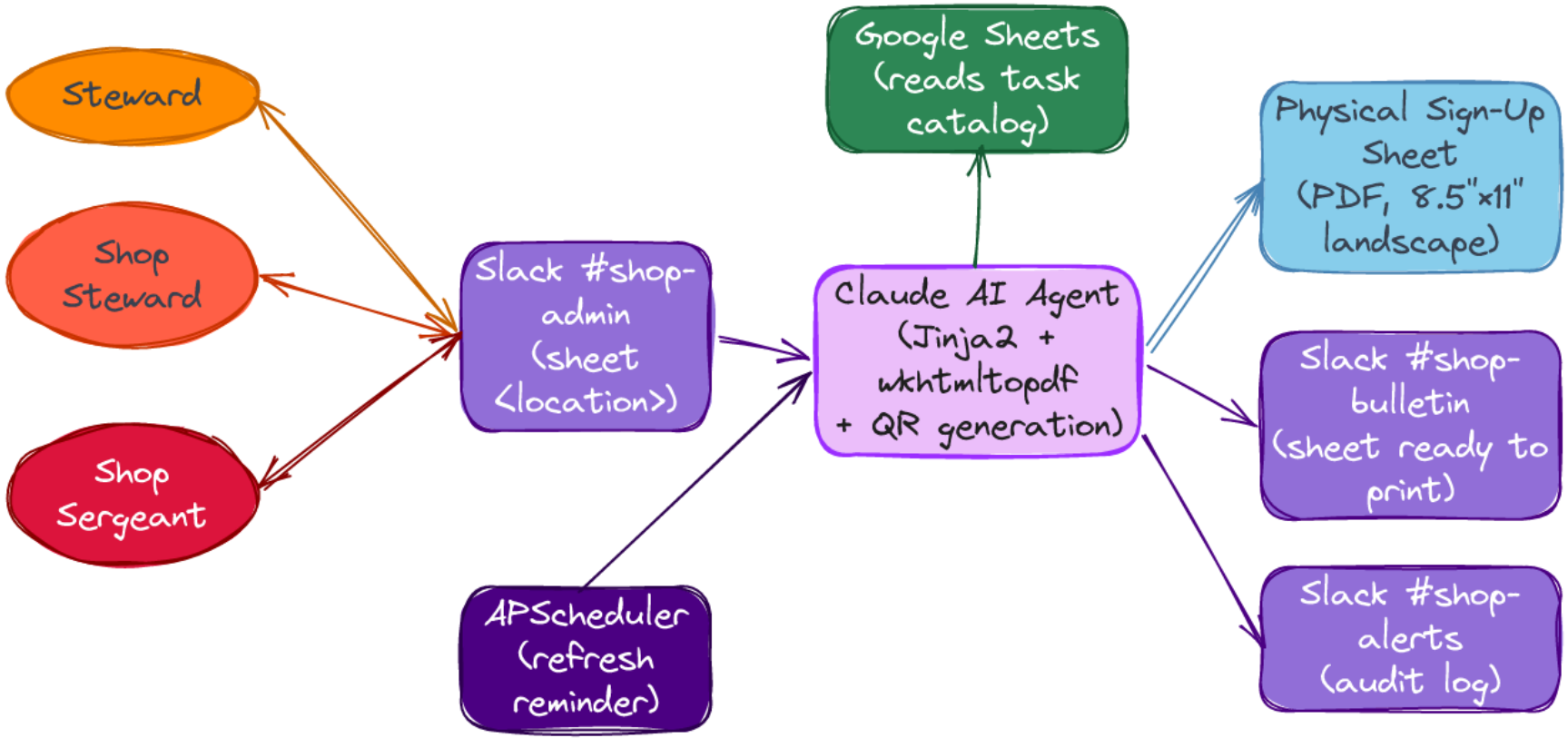
Task Sheet Pipeline



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Task Capture Pipeline

Records task completions from Members via

Method 1 - Single-Sheet QR → mobile web form

Method 2 - Slack Update in `#shop-queries`

Method 1: QR

Single-Sheet QR Flow

1. Member scans QR code on sign-up sheet with phone camera
2. Browser opens mobile web form automatically (no app, no login)
3. Member selects task + taps Submit **(≤ 3 taps total)**
4. Agentic Workers writes completion record to Google Sheets
5. Confirmation posted to `#shop-log`

Record written includes:

- Member name · Task ID · Completion date · Source: `QR-form`

Method 2: Slack

Slack Update Flow

1. Member posts in `#shop-queries` — free-form text or `task-done <task_id>`
2. Bot identifies task via NL processing (Claude AI) or task list selection
3. Bot confirms task identity with member before writing
4. Agentic Workers writes completion record to Google Sheets
5. Confirmation posted to `#shop-log`

Record written includes:

- Member name · Task ID · Completion date · Time Spent · Source: `Slack`

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Task Capture Pipeline — Key Requirements

#	Requirement
TC-1	Every completion record must include: Member name, Task ID, completion date, and capture method (`QR-form` or `Slack`)
TC-2	Completion records must be written to Google Sheets immediately upon submission
TC-3	A confirmation message must appear in `#shop-log` for every logged completion
TC-4	`#shop-alerts` must be notified whenever a capture event is processed
TC-5	The Method 1 (QR) capture flow must require ≤ 3 taps on a mobile device from QR scan to confirmation
TC-6	Method 1 (QR) must not require Members to create an account or install an app
TC-7	The bot must accept task completions in `#shop-queries` via free-form NL text or the `task-done <task_id>` command
TC-8	When no Task ID is provided, the bot must identify the task via NL processing or present a task list for selection
TC-9	The bot must confirm the identified task with the member before writing the completion record

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Task Capture Sign-Up Sheet — Example Output

Location:
Metalshop
Steward: Brad Hess

Volunteer Opportunities

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Task Name	Last Done	Frequency	Member Name / Date	Member Name / Date	Member Name / Date
Dust & Wipe Down Machine Do not use any cleaning solvant. Just a dry or damp cloth should be used. MSL-METAL-001	NA	Weekly			
Clear Off & Organize Tabletops MSL-METAL-002	NA	Daily			
Remove Anything That is Not CNC and Put Away Every CNC / Big Red thing has its place, so place it there. MSL-METAL-003	NA	Daily			
Vaccum Floor MSL-METAL-004	NA	Daily			
Check Dust Collection and Empty if Needed Pull the handle on the left side and remove the basket. Dump its contents in the dumpster in the back of the building. MSL-METAL-005	NA	Weekly			
Empty Shop Vac Dump its contents in the dumpster in the back of the building. MSL-METAL-006	NA	Bi-Weekly			
Empty Saw Dust On the front of the machine, remove the small basket and put in the dumpster. MSL-METAL-007	NA	Bi-Weekly			

Questions concerning the design & use of this form should be sent to Jeff Irland (xxx)

Metalshop sign-up sheet (v2) — single QR code in header · 8.5×11" landscape · black-and-white printable

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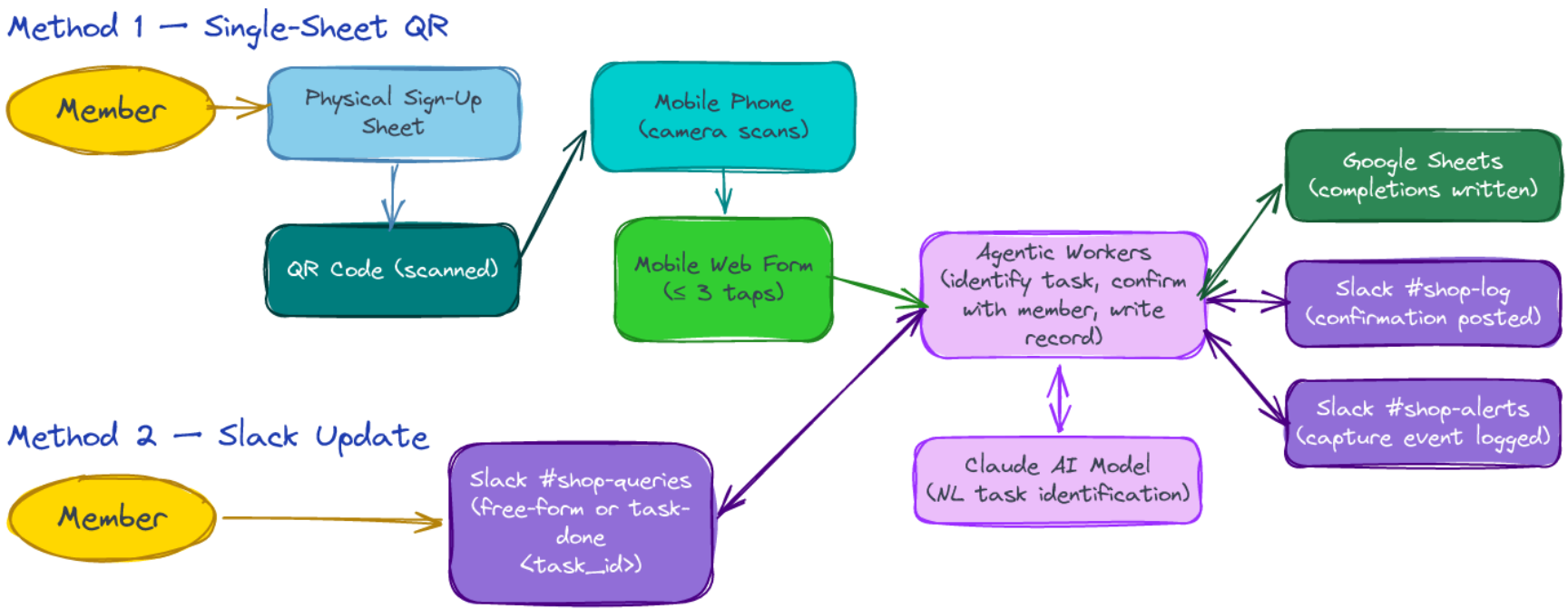
Task Capture Pipeline



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Task Reporting Pipeline

On-Demand Reports (NL query in Slack)

Report	Description	Access
Member	Tasks completed; compliance status	Own: any member · Other: Steward+
Location	Task status; overdue tasks	All Members (public)
Shop	Aggregate; top contributors	All Members (public)
Compliance	Who met 2hr/month requirement	Shop Steward + Sergeant only

Scheduled Reports (APScheduler)

- Weekly → `#shop-bulletin` : errors logged, location availability
- Monthly → `#shop-bulletin` : compliance summary
- Monthly → `#shop-bulletin` : top-contributor recognition post

Slack Channel Structure

Channel	Access	Purpose
`#shop-log`	All (read)	Task completion confirmations
`#shop-bulletin`	All (read)	Announcements + scheduled reports
`#shop-queries`	All	NL queries — public data
`#shop-admin` 🔒	Stewards+	Admin commands; restricted reports
`#shop-alerts` 🔒	Sgt only	Full audit trail + bot errors

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Task Reporting Pipeline — Key Requirements

#	Requirement
TR-1	Members may request reports via natural language in `#shop-queries` (public data only)
TR-2	Stewards, Shop Stewards, and Shop Sergeants may request any report type via `#shop-admin`
TR-3	The bot must respond to report requests inline in Slack
TR-4	`#shop-alerts` must be notified when any report is generated
TR-5	A monthly summary report must be automatically posted to `#shop-bulletin`
TR-6	A monthly top-contributors report must be automatically posted to `#shop-bulletin`
TR-7	A weekly report of errors logged, availability across all locations posted to `#shop-bulletin`

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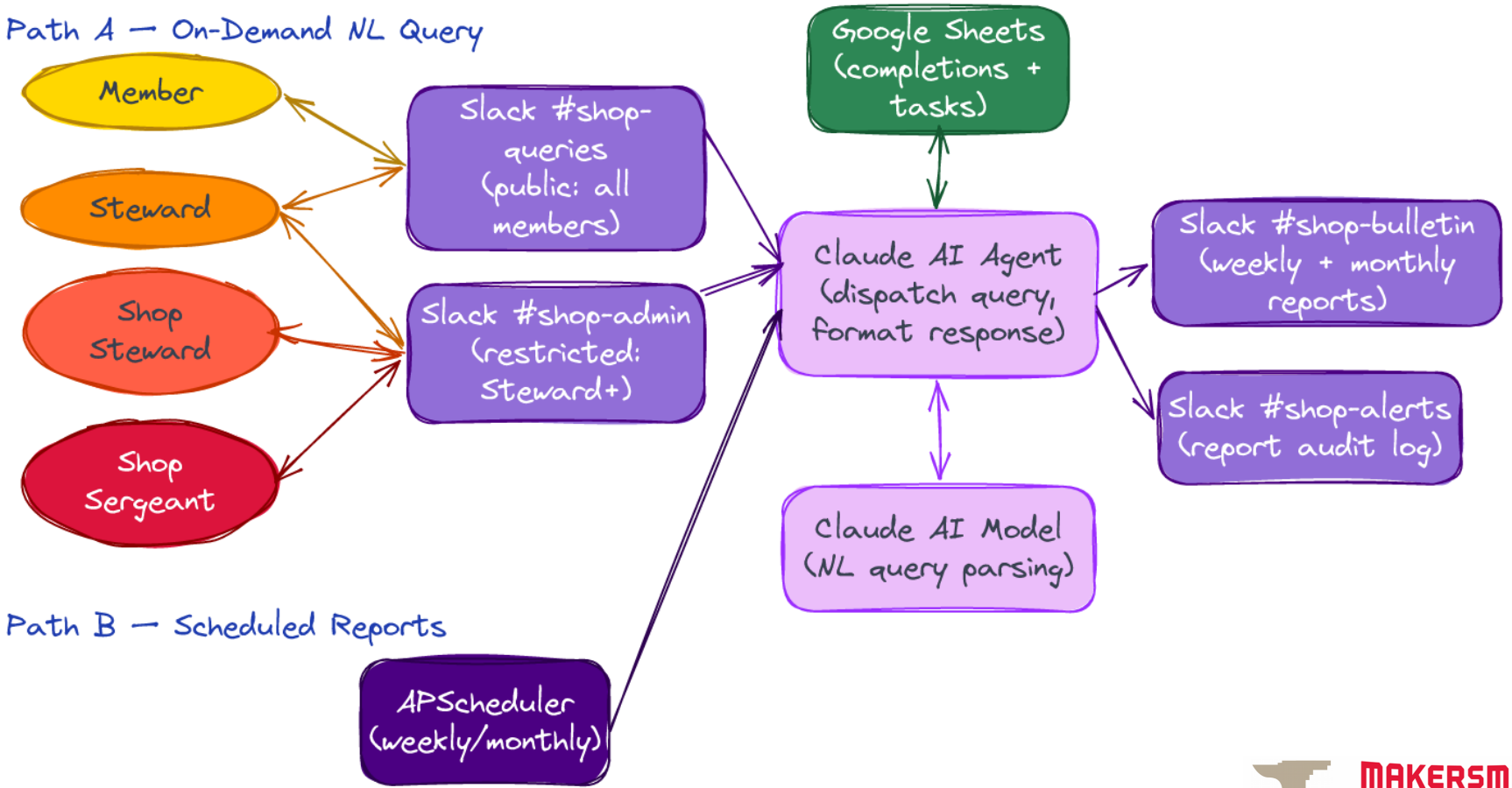
Task Reporting Pipeline



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Permission Model

Action	Member	Steward	Shop Steward	Shop Sergeant	Channel
Log task completion (QR or Slack)	✅	✅	✅	✅	web form / `#shop-queries`
Query own task history	✅	✅	✅	✅	`#shop-queries`
Query any Member's task history	❌	❌	✅	✅	`#shop-admin`
Query open tasks / location status	✅	✅	✅	✅	`#shop-queries`
Add / update / delete a task	❌	✅ own	✅	✅	`#shop-admin`
Generate / print sign-up sheet	❌	✅ own	✅	✅	`#shop-admin`
Request location or shop report	✅ public	✅	✅	✅	`#shop-queries` / `#shop-admin`
Trigger overdue reminder blast	❌	❌	✅	✅	`#shop-admin`
View bot errors / audit log	❌	❌	✅	✅	`#shop-alerts`
Suspend / restore — location scope	❌	✅	✅	✅	`#shop-admin`
Suspend / restore — shop scope	❌	❌	❌	✅	`#shop-admin`

> Steward permissions are scoped to their own location. Stewards CAN suspend/restore their location but CANNOT query other members' history or trigger reminder blasts.

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Non-Functional Key Requirements

Member Experience (Core Principle)

- ≤ 3 taps from QR scan to confirmation (Method 1)
- No account, no login, no app required for Method 1
- Paper fallback **always** available alongside digital

Record Retention

- Completion records retained **indefinitely** in Google Sheets
- Every record: member name + Task ID + date + capture method
- Exportable in CSV / Excel for 501©(3) compliance documentation

Availability & Resilience

- Google Sheets is durable — bot outage = zero data loss
- Physical sign-up sheet is the permanent manual fallback
- Two independent capture paths reduce single-point-of-failure risk

Access Control

- Slack channel membership is the primary access control mechanism
- No separate authentication system required
- Bot verifies actor role before executing restricted actions

Multi-Shop Extensibility

- Data model supports multiple shops from day one (MSL, MSP)
- All Task IDs are shop-prefixed (`MSL-` , `MSP-`) to prevent collisions
- Initial release: MSL only — adding MSP requires **no architectural changes**

Technology Stack

- Python 3.11+ · Slack Bolt · Google Sheets API v4
- Claude API (`claude-sonnet-4-6`) · Jinja2 + wkhtmltopdf
- `qrcode[pil]` · APScheduler · pytest

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Success Criteria — Member-Facing KPIs

ID	Metric	Definition	Target
M1	Member Compliance Rate	% of active Members who log ≥ 1 completion in a given month	≥ 50% monthly
M2	Task Completion Rate	% of posted tasks completed within their scheduled period	≥ 90%
M3	Data Completeness Rate	% of completion records with all 3 required fields (name + Task ID + date)	≥ 95%

Data gaps identified:

- **M1 — denominator gap:** No member registry exists. **Recommendation:** Add a `members` sheet to Google Sheets with a `role` column — provides the denominator for M1 and enables the compliance report.
- **M2 and M3** — Fully computable from existing `tasks` and `completions` sheets. No gaps.

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Success Criteria — Operational KPIs

ID	Metric	Definition	Target
O1	Steward Adoption Rate	% of Stewards who issued ≥ 1 bot command in the period	100% within 60 days of launch
O2	Sheet Refresh Latency	Time from Slack command to PDF available for download	≤ 24 hours
O3	Overdue Task Alert Rate	% of overdue tasks for which an APScheduler reminder was sent	100%
O4	Top Contributor Recognition	Monthly top-contributor report posted to `#shop-bulletin` on time	100% monthly

Data gaps identified:

- **O1 — denominator gap:** Same as M1: no steward registry. Fixed by the `members` sheet with a `role` column.
- **O2 — scope:** Physical posting to wall is not observable. O2 is redefined as "Slack command → PDF ready."
- **O3 — log requirement:** Bot must log each overdue-reminder event with `task_id` + `alert_timestamp`. **Must be added to TR requirements.**

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Out of Scope

Item	Notes
Makersmiths Purcellville (MSP)	Future release. Architecture must accommodate MSP without structural changes, but no MSP deployment is planned.
Capture methods not selected	Once Phase 0 trial picks a method, the other is documented for reference only.
Email / SMS notifications	Slack is the sole notification channel. No alternatives planned.
Native mobile app	No native iOS/Android app, mobile experience delivered via mobile-optimized web form.
Real-time dashboards	Reports delivered on-demand via Slack or on a scheduled basis. NL queries cover real-time needs.

Open Questions

Before implementation begins, these deserve deliberate discussion:

- **Permission model edges** — Stewards can suspend their location (reasonable for emergencies) but can't trigger overdue reminders for their own location. Both edges worth discussing with actual stewards before the permission model is coded.
- **Track task duration?** — The 2hr/month requirement tracks task *count*, not actual *hours*. Adding an estimated `time` field per task enables true compliance checking. Better to add to the schema now than retrofit later.
- **Paper fallback sunset?** — If Method 1 proves reliable and shop Wi-Fi is consistently available, the permanent paper fallback adds complexity without sufficient benefit. Should it have a planned deprecation trigger?
- **APScheduler or Claude Routines?** — APScheduler runs in-process; jobs are silently skipped on bot restart. Claude Routines are managed cloud-scheduled — no persistent process needed, but adds per-run token cost. A hybrid is possible.
- **Keep both capture methods post-trial?** — If Phase 0 confirms Method 1 works reliably, retiring Method 2 removes design scope and developer ambiguity. The trial data should drive this decision.

Questions & Discussion

- eMail: jeff.irland@verison.net
- Slack: jeff.irland
- Source: `github.com/jeffskinnerbox/makersmiths-shop`
  - Full requirements: `docs/requirements.md`
  - Next document: `docs/specifications.md`

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